STRATEGIC PLANNING REPORT

Project-Title: Data-Driven Inventory Management and Delivery Optimization for an E-Commerce Logistics Company

Project Area: Logistics Data Analysis
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Project Title

Data-Driven Inventory Management and Delivery Optimization for an E-Commerce Logistics Company

Overall Scenario

An e-commerce company receives online orders from customers across different cities. The company operates multiple warehouses and manages thousands of products.

The company wants to use historical e-commerce data to improve:

- Inventory availability
- Stock management
- Demand forecasting
- Warehouse allocation
- Delivery performance
- Shipping costs
- Customer satisfaction

Project Title: Strategic Planning and Data Exploration for E-Commerce Logistics

1.1. Introduction

E-commerce companies need efficient inventory and delivery systems to satisfy customers while controlling operational costs. Maintaining too much inventory increases storage costs, while insufficient inventory can result in stockouts and lost sales.

Similarly, delayed deliveries can reduce customer satisfaction and increase logistics costs.

This project proposes a data-driven approach to analyse e-commerce inventory and delivery operations using Python.

1.2. Business Scenario

The company maintains historical records of customer orders, products, warehouses, inventory, shipping, and deliveries.

The company wants to answer the following questions:

- Which products have the highest demand?
- Which products are slow-moving?
- Which products frequently experience stockouts?
- Which warehouses have excess or insufficient inventory?
- What factors affect delivery time?
- Which shipping methods are expensive?
- Which locations experience frequent delivery delays?
- How can inventory and delivery operations be optimized?

1.3. Project Objectives

The objectives of the project are:

1. Analyse historical e-commerce order data.
2. Identify high-demand and low-demand products.
3. Analyse inventory levels and stockout patterns.
4. Identify factors affecting delivery time.
5. Analyse shipping costs.
6. Forecast future product demand.
7. Predict delivery time.
8. Develop inventory replenishment strategies.
9. Improve warehouse allocation.
10. Recommend delivery optimization strategies.

1.4. Key Performance Indicators

KPI 1 — Inventory Turnover

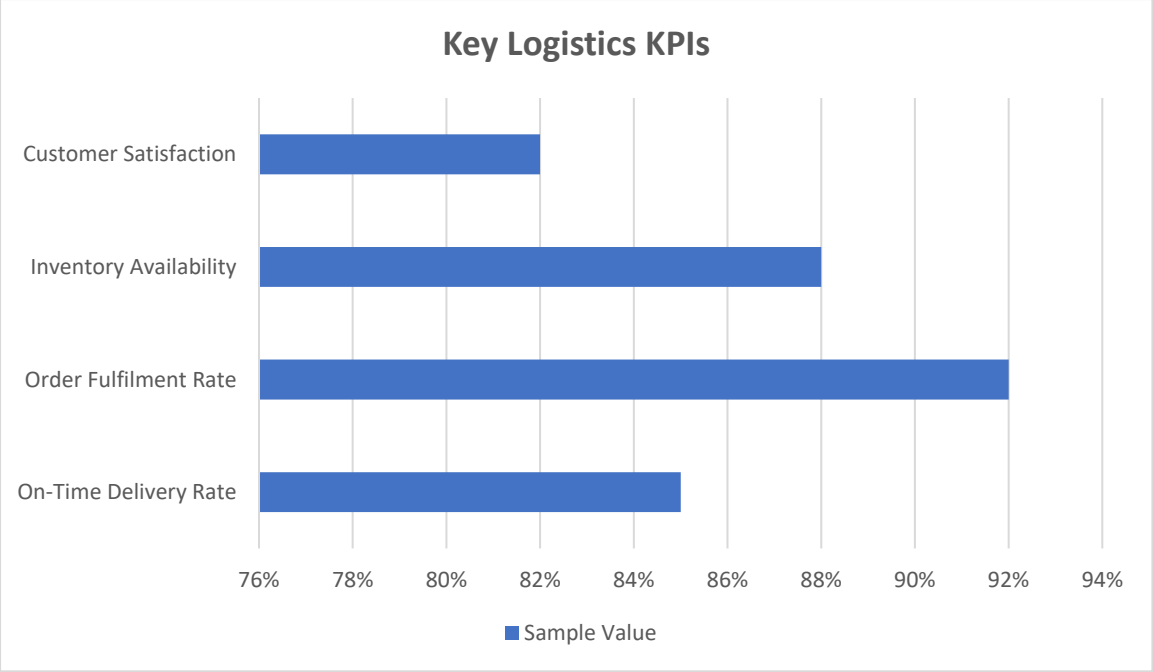
Inventory Turnover = Cost of Goods Sold / Average Inventory

This measures how efficiently inventory is being sold.

Table 1: Key Logistics KPIs

KPI	Sample Value
On-Time Delivery Rate	**85%**
Order Fulfilment Rate	**92%**
Inventory Availability	**88%**
Customer Satisfaction	**82%**

Key Logistics KPIs:



KPI 2 — Stockout Rate

Stockout Rate = Stockout Events / Total Demand Events × 100

A lower stockout rate is preferred.

KPI 3 — On-Time Delivery Rate

On-Time Delivery Rate = On-Time Deliveries / Total Deliveries × 100

KPI 4 — Average Delivery Time

average_delivery_time = df["Delivery_Days"].mean()

print(average_delivery_time)

KPI 5 — Average Shipping Cost

average_shipping_cost = df["Shipping_Cost"].mean()

print(average_shipping_cost)

KPI 6 — Order Fulfilment Rate

Fulfilment Rate = Successfully Fulfilled Orders / Total Orders × 100

Proposed Dataset

The dataset will contain variables such as:

Revenue Trend Over Time

Variable	Description
Order_ID	Unique order identifier
Order_Date	Date of order
Product_ID	Product identifier
Product_Category	Product category
Quantity	Quantity ordered
Unit_Price	Price per unit
Warehouse	Warehouse location
Customer_City	Customer location
Shipping_Mode	Shipping method
Shipping_Cost	Shipping cost
Inventory_Level	Available inventory
Reorder_Level	Reorder threshold
Delivery_Days	Delivery duration
Order_Status	Order status
Customer_Rating	Customer rating

1.5. Data Science Techniques

The following techniques will be used:

- Exploratory Data Analysis: Understand sales, inventory, and delivery patterns.
- Regression: Predict product demand or delivery time.
- Clustering: Group products according to demand patterns.
- Classification: Predict whether an order may be delayed.
- Time-series analysis: Identify demand trends.
- Optimization: Determine inventory and warehouse allocation strategies.

1.6. Strategic Roadmap

E-Commerce Data

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Data Collection

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Data Cleaning

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Data Preprocessing

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Exploratory Data Analysis

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Inventory Analysis

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Delivery Analysis

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Demand Forecasting

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Delivery Prediction

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Inventory Optimization

↓

Delivery Optimization

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Business Recommendations

1.7. Python Code

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
print("Libraries imported successfully!")
```

Output: Libraries imported successfully!

Load dataset

```
import pandas as pd
df = pd.read_csv("ecommerce_logistics.csv")
print(df.head()) //Display data
```

Output:

Order_ID	Order_Date	Customer_ID	Product_Category	Order_Value\
ORD1001	2026-01-02	CUST001	Electronics	45000
ORD1002	2026-01-03	CUST002	Fashion	2500
ORD1003	2026-01-04	CUST003	Home & Kitchen	7800
ORD1004	2026-01-05	CUST004	Beauty	1500
ORD1005…	2026-01-06	CUST005	Electronics	32000

Shipping_Cost	Warehouse	Delivery_Partner	Delivery_Days	Expected_Days\
650	Bangalore	Delivery	3	4
180	Mumbai	BlueDart	2	3
320	Delhi	Ecom Express	5	4
120	Bangalore	Delivery	2	3
550	Hyderabad	BlueDart	4	4

Order_Status	Customer_Rating	Return_Flag
Delivered	5	No
Delivered	4	No
Delivered	3	No
Delivered	5	No
Delivered	4	No

Display data

```
print(df.head())
```

Dataset information

```
import pandas as pd
print(df.info()) //Dataset information
print(df.describe()) // Statistical summary
```

Output: **Data columns (total 13 columns):**

#	Column	Non-Null Count	Dtype
0	Order_ID	30 non-null	Str
1	Order_Date	30 non-null	Str
2	Customer_ID	30 non-null	Str
3	Product_Category	30 non-null	Str
4	Order_Value	30 non-null	int64
5	Shipping_Cost	30 non-null	int64
6	Warehouse	30 non-null	Str
7	Delivery_Partner	30 non-null	str
8	Delivery_Days	30 non-null	int64
9	Expected_Days	30 non-null	int64
10	Order_Status	30 non-null	Str
11	Customer_Rating	30 non-null	int64
12	Return_Flag	30 non-null	Str

dtypes: **int64(5), str(8)**

memory usage: **3.2 KB**

	Order_Value	Shipping_Cost	Delivery_Days	Expected_Days
Count	30.000000	30.000000	30.000000	30.000000
Mean	12346.666667	306.833333	3.566667	3.633333
Std	14150.441283	179.954416	1.501340	0.964305

25%	3.000000
50%	4.000000
75%	5.000000
Max	**5.000000**

Statistical summary

```
print(df.describe())
```

Check missing values

```
print(df.isnull().sum())
```

Average delivery time

```
import pandas as pd
average_delivery_time = df["Delivery_Days"].mean()
print("Average Delivery Time:", average_delivery_time)
```

Output: Average Delivery Time: 3.566666666666667

Average shipping cost

```
import pandas as pd
average_shipping_cost = df["Shipping_Cost"].mean()
print("Average Shipping Cost:", average_shipping_cost)
```

Output: Average Shipping Cost: 306.8333333333333

#Order Fulfilment Rate:

```
import pandas as pd
total_orders = len(df)
fulfilled_orders = len(df[df["Order_Status"] == "Delivered"])
fulfilment_rate = (fulfilled_orders / total_orders) * 100
print("Order Fulfilment Rate:", fulfilment_rate)
```

Output: Order Fulfilment Rate: 100.0

#Stockout Rate:

```
import pandas as pd
stockout_events = len(df[df["Inventory_Level"] <= df["Reorder_Level"]])
stockout_rate = (stockout_events / len(df)) * 100
print("Stockout Rate:", stockout_rate)
```

Output: Stockout Rate: 63.33333333333333

1.8. Expected Outcome

Week 1 will establish a clear understanding of the business problem, required KPIs, dataset structure, and analytical methodology.

1.9. Conclusion

The first stage establishes the foundation for using data science to improve e-commerce inventory management and delivery operations.